Regression Metrics

MAE

MSE

RMSE

R2 Score

Adjusted R2 score

Regression Metrics

Thursday, May 13, 2021 11:56 AM

1) MAE

2) MSE

3) RMSE

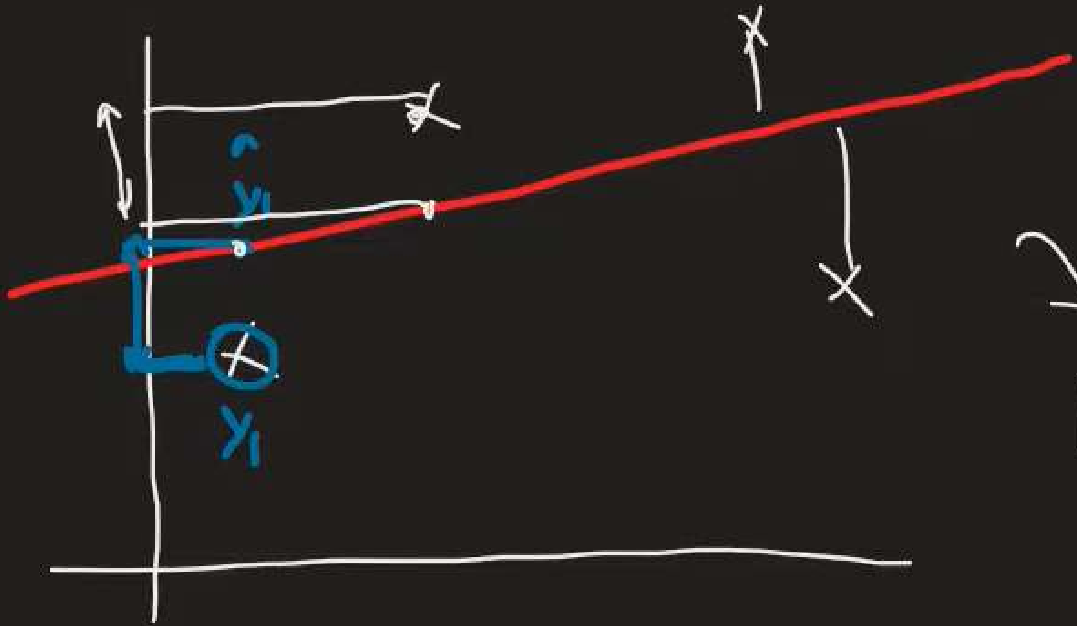
4) R2 score

5) Adjusted R2 score

MAE

Thursday, May 13, 2021 11:56 AM

x | y



$$\frac{|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| + \dots + |y_n - \hat{y}_n|}{n}$$

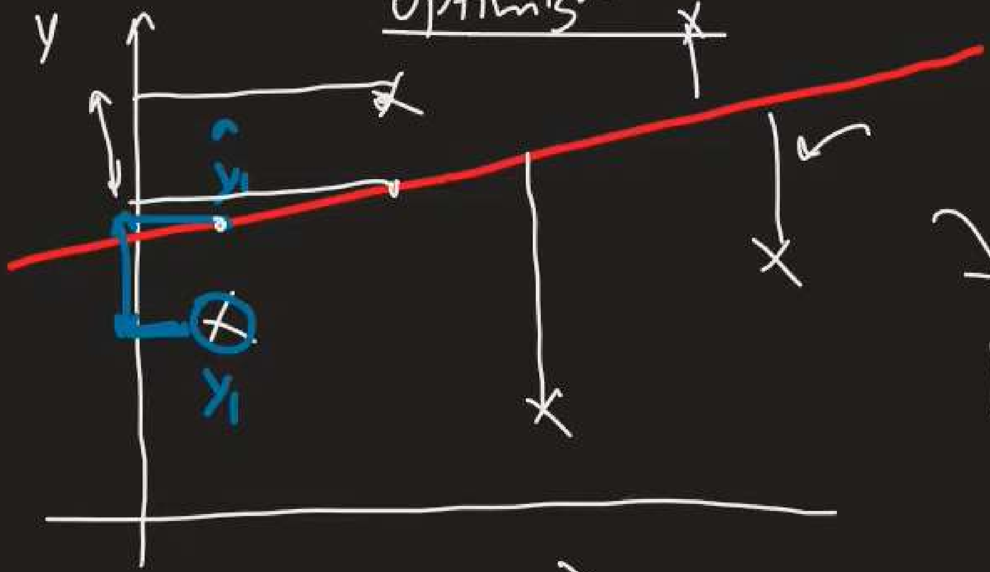
$$\boxed{mae = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}}$$

MAE

Thursday, May 13, 2021 11:56 AM

cgpa x | y package (lpa)

Optimization



$$\frac{|y_1-\hat{y}_1| + |y_2-\hat{y}_2| + \dots + |y_n-\hat{y}_n|}{n}$$

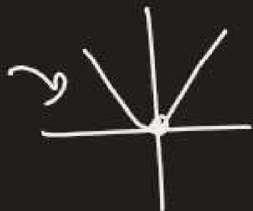
1.5 lpa

$$\text{mae} = \frac{\sum_{i=1}^{n} |y_i-\hat{y}_i|}{n}$$

Advantage

1) same unit
2) Robust outliers

Disadvantage

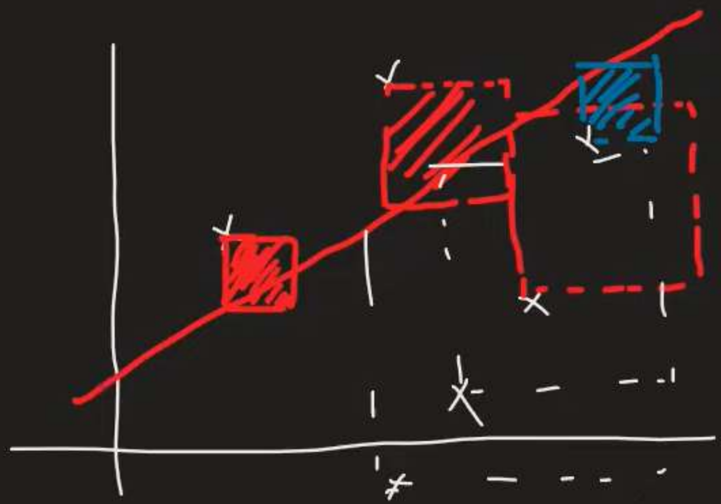


MSE

Thursday, May 13, 2021 11:56 AM

mean squared error

cgpa X | y package (lpa)



11.25

$$mae = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$

mse → L function

$(y_i - \hat{y}_i)^2$

Advantage → L function

Disadvantage → Robust to outliers

$$mse = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$

y_lpa

mse − $(lpa)^2$

RMSE

Thursday, May 13, 2021 11:56 AM

$$Rmse = \sqrt{mse}$$

mse

$$= \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

rmse - lpa

y - lpa

benefit

disadvantage

Robust
outliers

R2 Score

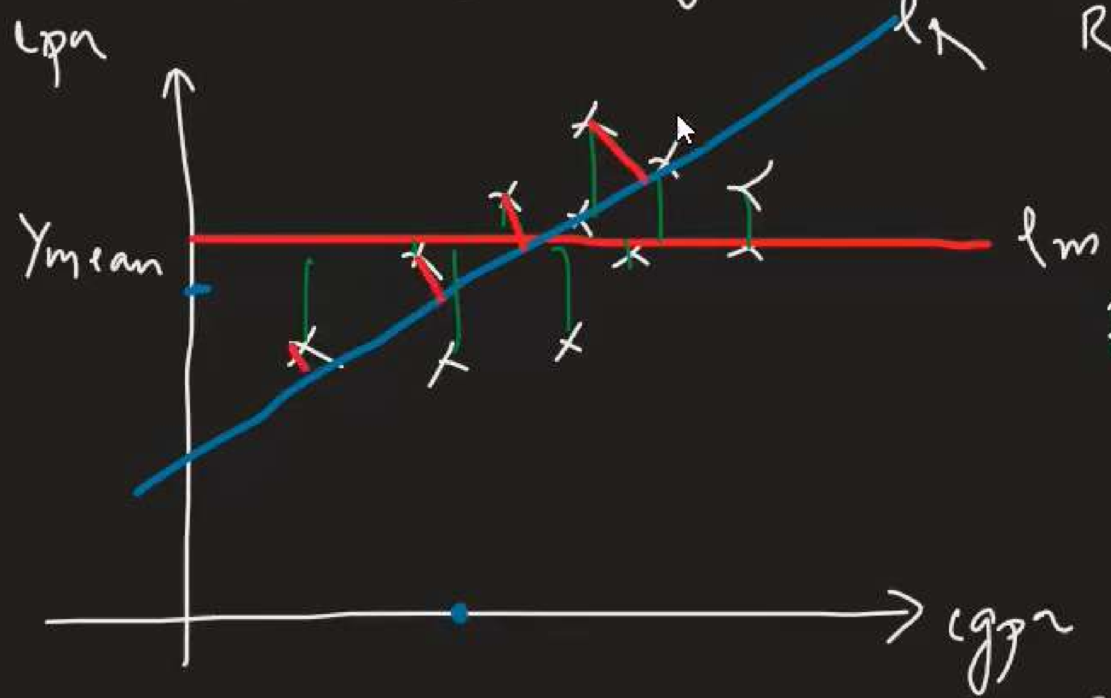
Thursday, May 13, 2021 11:56 AM

$\boxed{100} \rightarrow$ mean $\rightarrow$ 3.4

cgpa | package (lpa)

coeff of dete

↳ goodness of fit



$$R2 = 1 - \frac{SS_R}{SS_m}$$

$= 1$ $\quad = 0$

$$R2 = 1 - \frac{\left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2\right]_{Reg}}{\left[\sum_{i=1}^{n} (y_i - \bar{y}_i)^2\right]_{m}}$$

$\rightarrow 0$

$R_2 = 0$

$R_2 = 1$

Sum of squared error Reg

$0 \rightarrow R2 \longrightarrow 1$

Adjusted R2 score

Thursday, May 13, 2021 11:57 AM

0.80 0.90

R2 score ↑

cgpa | lpa iq

→ temp

R2 ↑ _

Adjusted R2

$1 - \uparrow$ ↓ $[1 - \downarrow]$ ↑

R2 →

n → no. of rows

K = independent

K=1, K=2, K=3

$$\boxed{R^2_{adj}} = 1 - \left[\frac{\boxed{(1-R2)}\,(n-1)}{(n-1-K)}\right]$$

Multiple LR

temp

Adj R2 ↓

iq

R^2_{adj} ↑